ECE 343 - Chapter 04
Vector Analysis



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Lesson	Topics	Reading	Notes
2	Course Overview & Introduction	Ch.1	LO 1.1-1.3
	Chapter 2: Electric and Magnetic Fields		
3	Fields, Parameters, Field Quantities	Ch. 2	LO 2.1-2.4
	Chapter 3: Transmission Lines		
4	Lumped Element Model	3.1-3.4	LO 3.1
5	Telegrapher's Equations	3.5-3.6	LO 3.1
6	Characteristic Impedance & Wave Propagation	3.7-3.9	LO 3.2
7	Coaxial, Reflection Coefficient, VSWR	3.10,3.12-3.14	LO 3.2-3.3
8	Terminated Transmission Lines	3.15,3.17-3.19	LO 3.4
9	Impedance Matching	3.21-3.23	LO 3.4
10	Impedance Matching	3.21-3.23	LO 3.4
11	Smith Charts	Supplemental	LO 3.5
12	Smith Charts	Supplemental	LO 3.5
13	Smith Charts	Supplemental	LO 3.5
14	Smith Charts	Supplemental	LO 3.5
15	Transients		LO 3.6
16	Reflection/Transmission Coefficient Lab	Lab Packet	
	Chapter 4: Vector Analysis		
17	Vector & Coordinate System Review	Ch. 4	**GR 1 due**
	Chapter 5: Electrostatics		
18	Coulomb's Law	5.1-5.3	
19	Surface and Volume Charge	5.4	
20	Gauss' Law (Integral Form)	5.5-5.6	
21	Gauss' Law (Differential Form)	5.7-5.8	
22	Electric Potential	5.9-5.10, 5.15	

Learning Objectives

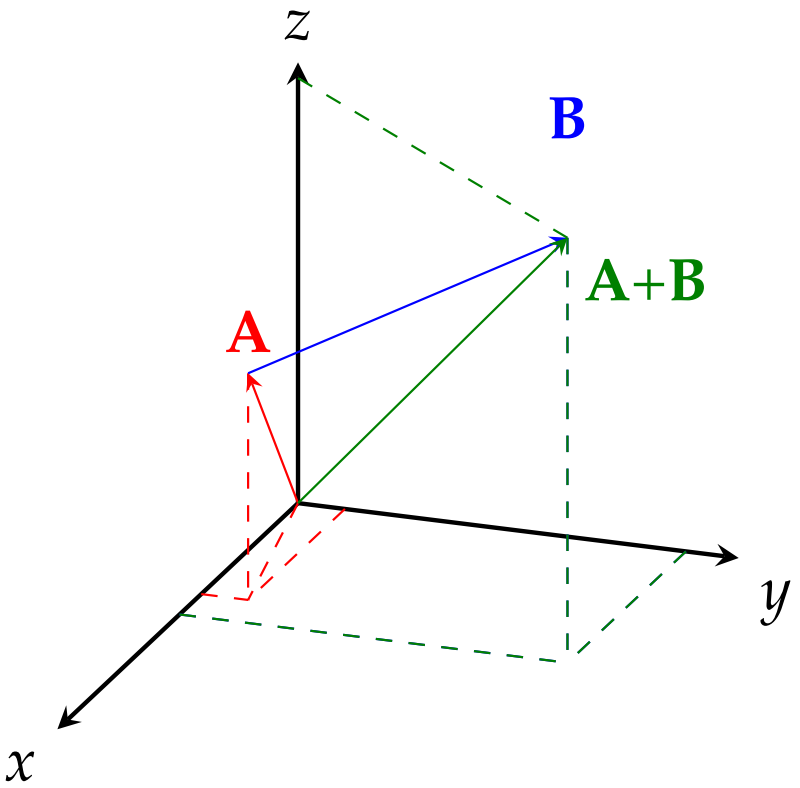
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3.1 Recognize and apply common vector calculus operations, such as: gradient, divergence, curl, and Laplacian.

3.2 Calculate surface and volume integrals in rectangular, cylindrical, and spherical coordinate systems.

3.3 Differentiate between Stoke's Theorem and the Divergence Theorem and apply them to various problems to simplify integrals.

VECTORS - REVIEW



$$\vec{A} = 0.6\hat{\mathbf{x}} + 0.1\hat{\mathbf{y}} + 0.5\hat{\mathbf{z}}$$

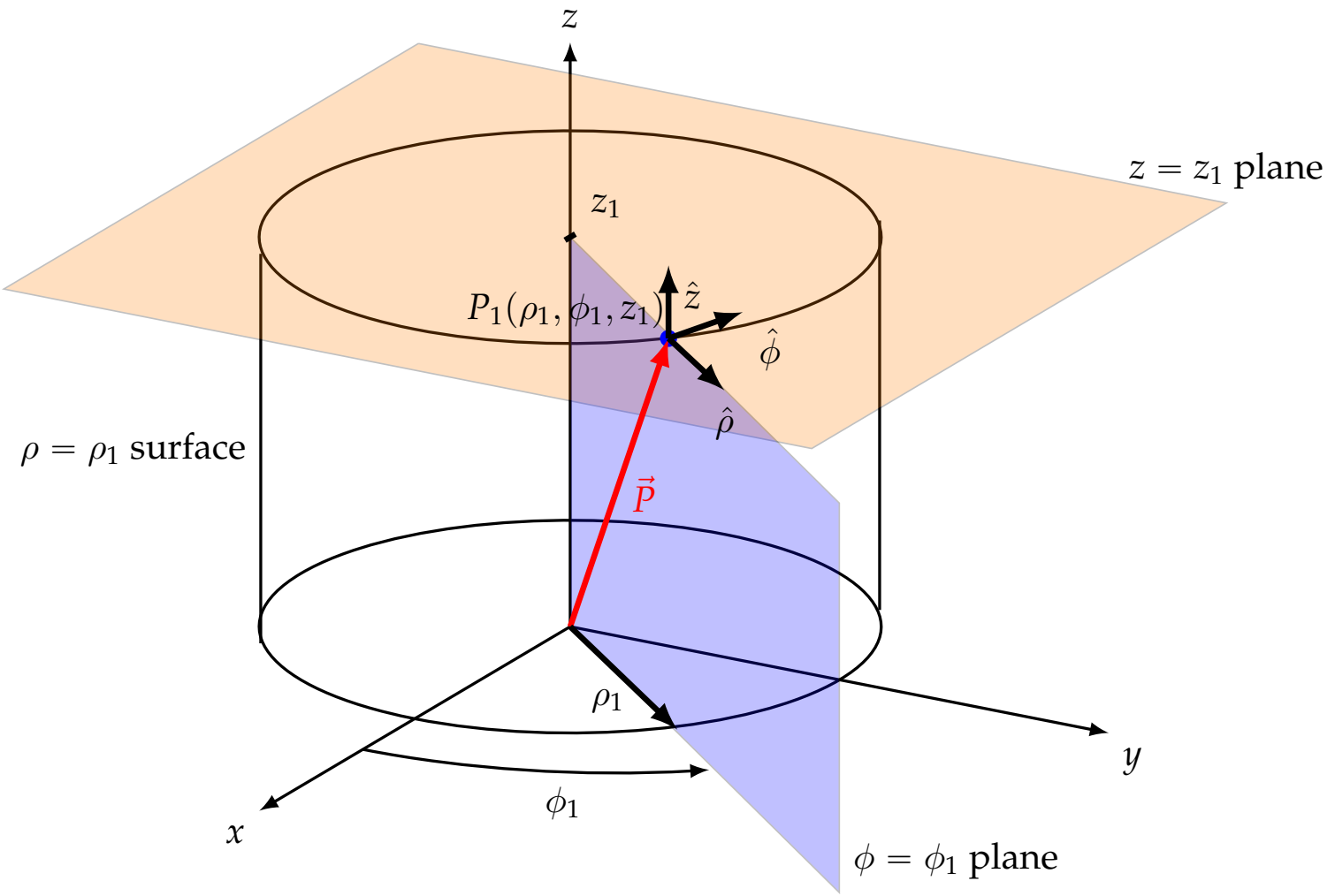
$$\vec{B} = 0.75\hat{\mathbf{x}} + 0.9\hat{\mathbf{y}} + \hat{\mathbf{z}}$$

$$\vec{A} + \vec{B} =$$

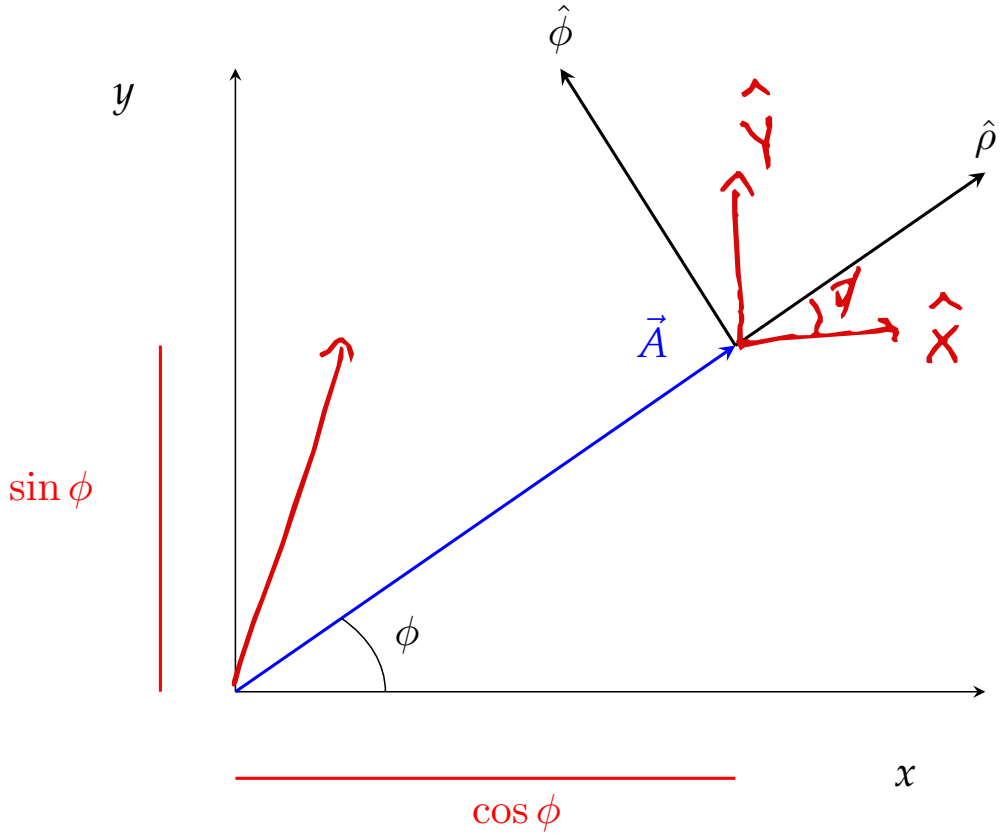
$$\vec{A} \cdot \vec{B} = |A||B|\cos\theta$$

$$\vec{A} \times \vec{B} =$$

Cylindrical Coords - Review



Cylindrical Coords - Transformation



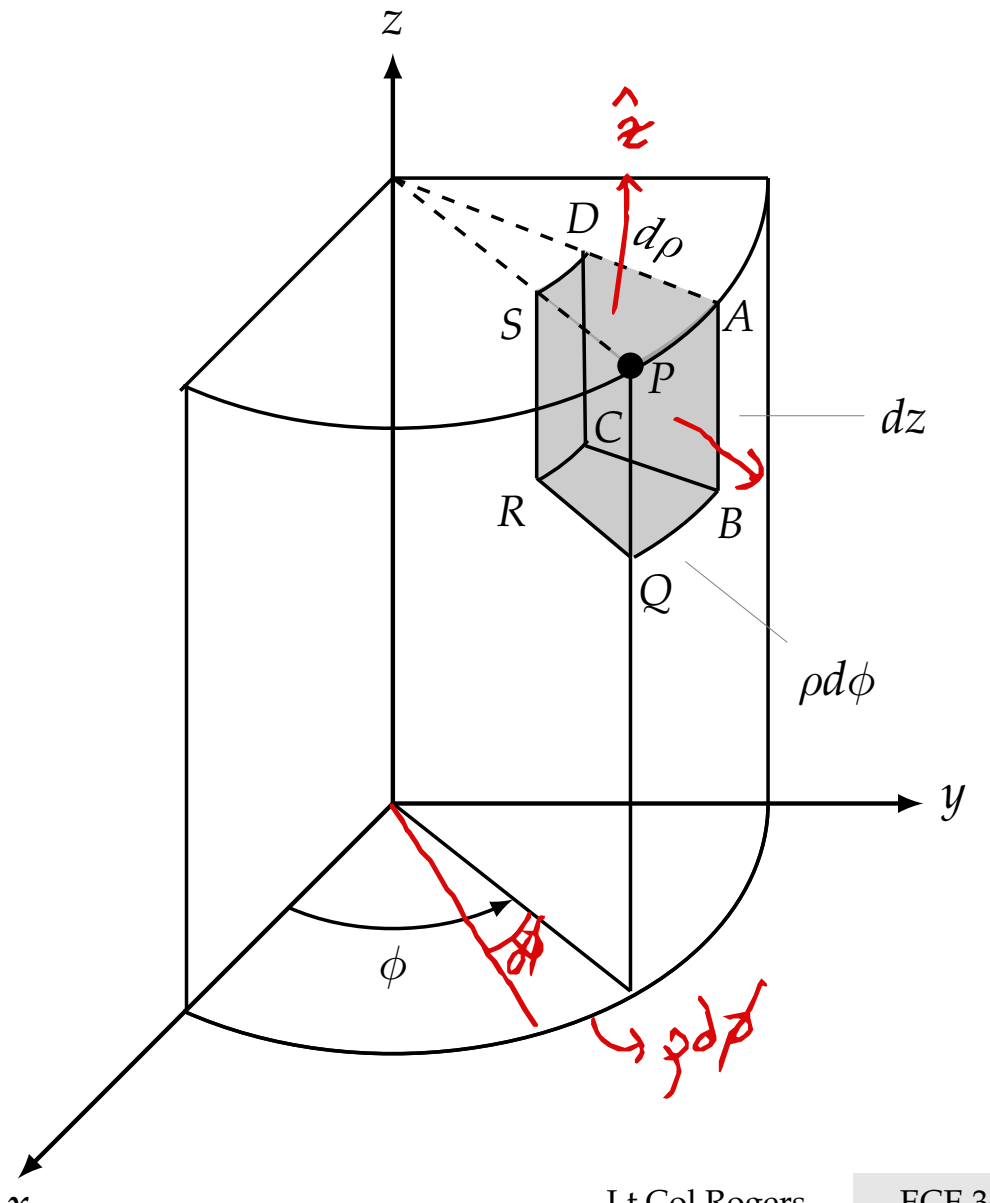
$$x = \rho\cos\phi$$
$$y = \rho\sin\phi$$
$$z = z$$

$$\rho = \sqrt{x^2 + y^2}$$
$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$
$$z = z$$

$$\hat{\boldsymbol{\rho}} = \hat{\mathbf{x}}\cos\phi + \hat{\mathbf{y}}\sin\phi$$
$$\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}}\sin\phi + \hat{\mathbf{y}}\cos\phi$$
$$\hat{\mathbf{z}} = \hat{\mathbf{z}}$$

$$\hat{\mathbf{x}} = \hat{\boldsymbol{\rho}}\cos\phi - \hat{\boldsymbol{\phi}}\sin\phi$$
$$\hat{\mathbf{y}} = \hat{\boldsymbol{\rho}}\sin\phi + \hat{\boldsymbol{\phi}}\cos\phi$$
$$\hat{\mathbf{z}} = \hat{\mathbf{z}}$$

Cylindrcial Differential Elements



Lengths:

$$d\vec{\ell} = \hat{\rho} d\ell_\rho + \hat{\phi} d\ell_\phi + \hat{z} d\ell_z$$

$$d\ell_\rho = d\rho$$

$$d\ell_\phi = \rho d\phi$$

$$d\ell_z = dz$$

Surfaces

$$d\vec{S}_z = \pm \hat{z}\, d\rho\, \rho d\phi = \rho d\rho d\phi$$

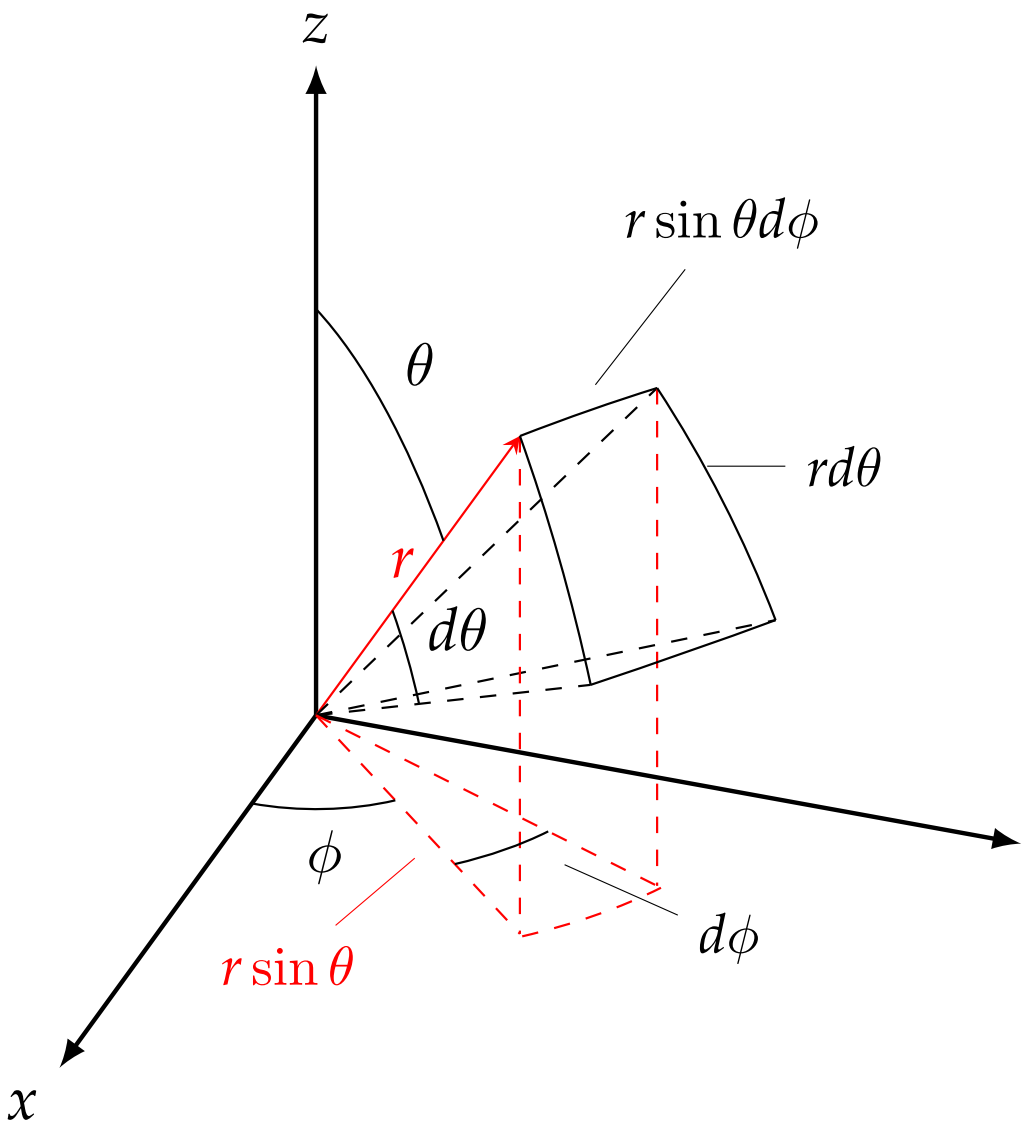
$$d\vec{S}_\rho = \pm \hat{\rho}\, \rho d\phi dz$$

$$d\vec{S}_\phi = \pm \hat{\phi}\, d\rho dz$$

Volume:

$$dV = \rho d\rho d\phi dz$$

Spherical Differential Elements



Lengths:

$$dl_r = dr$$
$$dl_\theta = r\,d\theta$$
$$dl_\phi = r\sin\theta\,d\phi$$

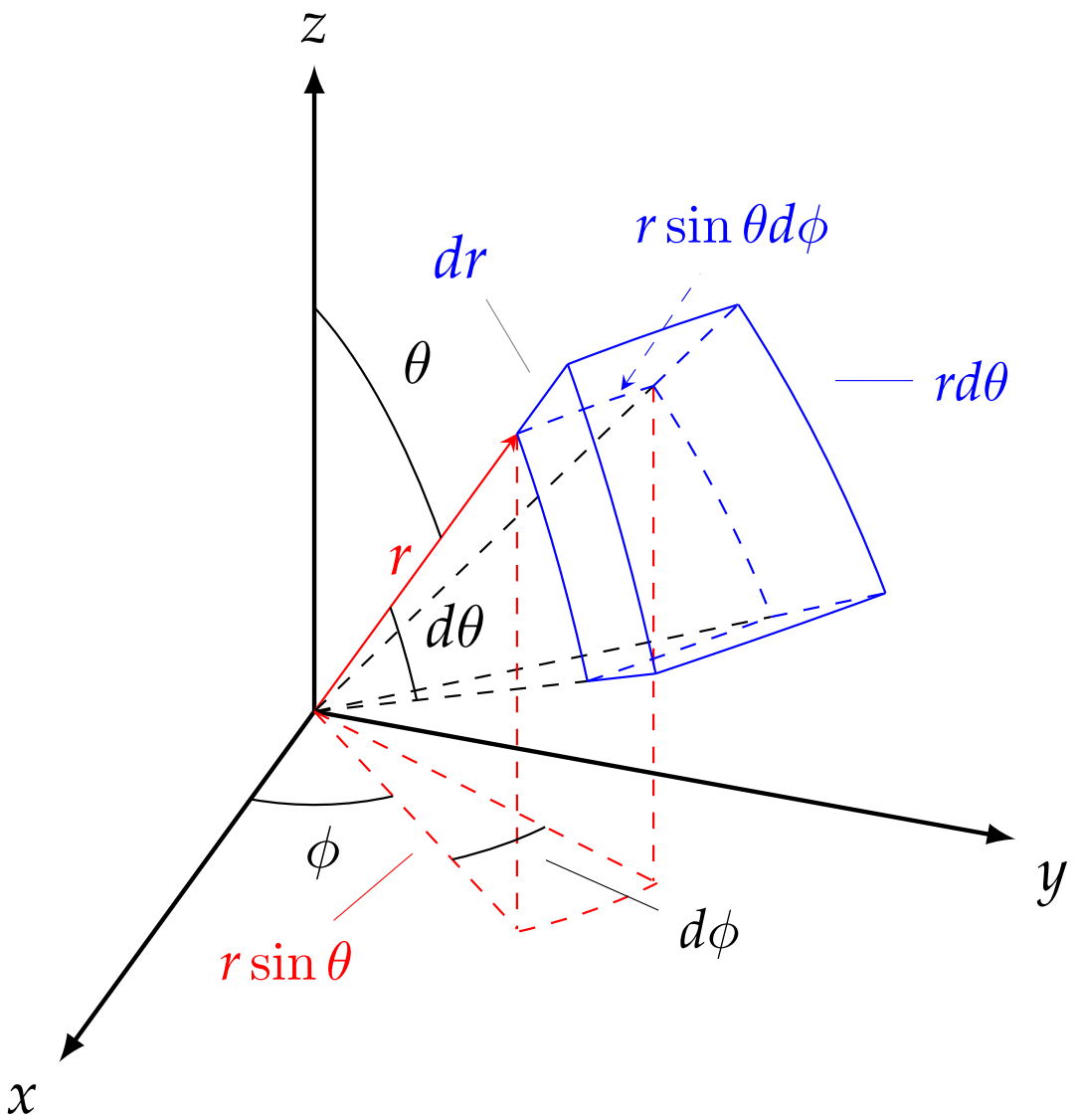
Surfaces:

$$dS_r = r^2\sin\theta\,d\theta\,d\phi$$
$$dS_\theta = r\sin\theta\,dr\,d\phi$$
$$dS_\phi = r\,dr\,d\theta$$

Volume

$$dV = r^2\sin\theta\,dr\,d\theta\,d\phi$$

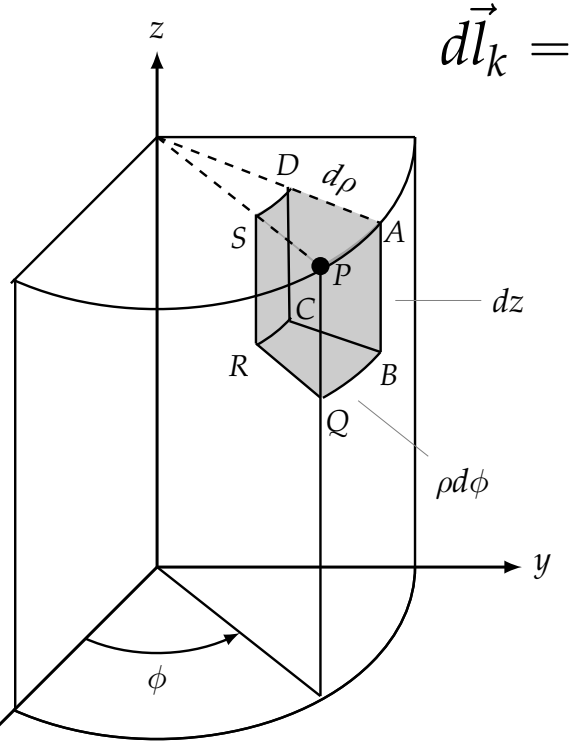
Spherical Differential Elements - Volume

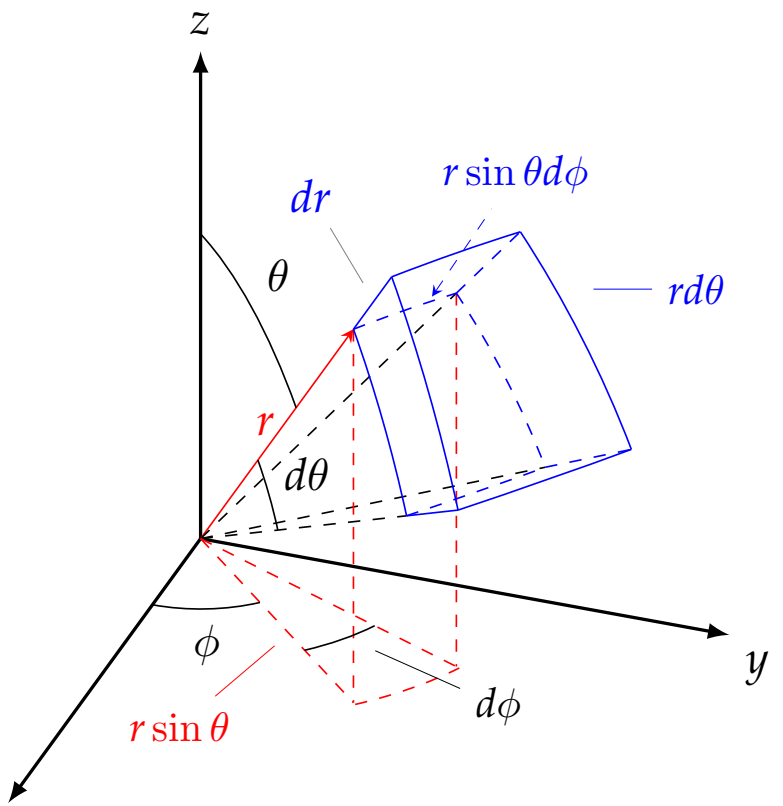


Generalized Coordinate System

$$\vec{A} = \hat{u}_1 A_1 + \hat{u}_2 A_2 + \hat{u}_3 A_3$$

$$d\vec{l}_k = \hat{u}_k h_k du_k$$





h_1

$d\vec{l}_1$ u_1 ⟶ $\hat{\rho}(1)d\rho$ $h_1 = 1$

$d\vec{l}_2$ u_2 ⟶ $\hat{\phi}\,\rho\,d\phi$ $h_2 = \rho$

$d\vec{l}_3$ u_3 ⟶ $\hat{z}(1)dz$ $h_3 = 1$

u_1 ⟶ $\hat{r}\,dr$ $h_1 = 1$

u_2 ⟶ $\hat{\theta}\,r\,d\theta$ $h_2 = r$

u_3 ⟶ $\hat{\phi}\,r\sin\theta\,d\phi$ $h_3 = r\sin\theta$

Nabla (del) Operator

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$$\nabla = \hat{u}_1 \frac{1}{h_1} \frac{\partial}{\partial u_1} + \hat{u}_2 \frac{1}{h_2} \frac{\partial}{\partial u_2} + \hat{u}_3 \frac{1}{h_3} \frac{\partial}{\partial u_3}$$

- Rectangular $\longrightarrow$
- Cylindrical $\longrightarrow$ $\hat{\rho} \frac{1}{1} \frac{\partial}{\partial \rho} + \hat{\phi} \frac{1}{\rho} \frac{\partial}{\partial \phi} + \hat{z} \frac{1}{1} \frac{\partial}{\partial z}$
- Spherical $\longrightarrow$ $\hat{r} \frac{1}{1} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{1}{r \sin\theta} \frac{\partial}{\partial \phi}$

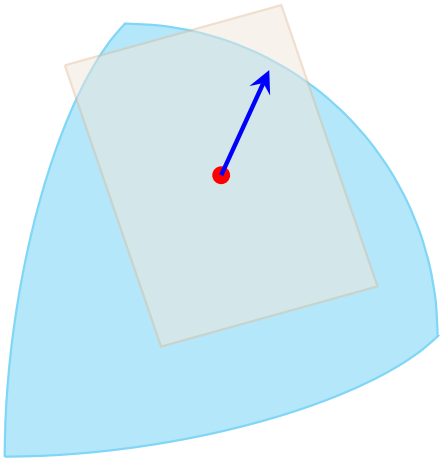
Gradient

$f(v_1, v_2, v_3)$

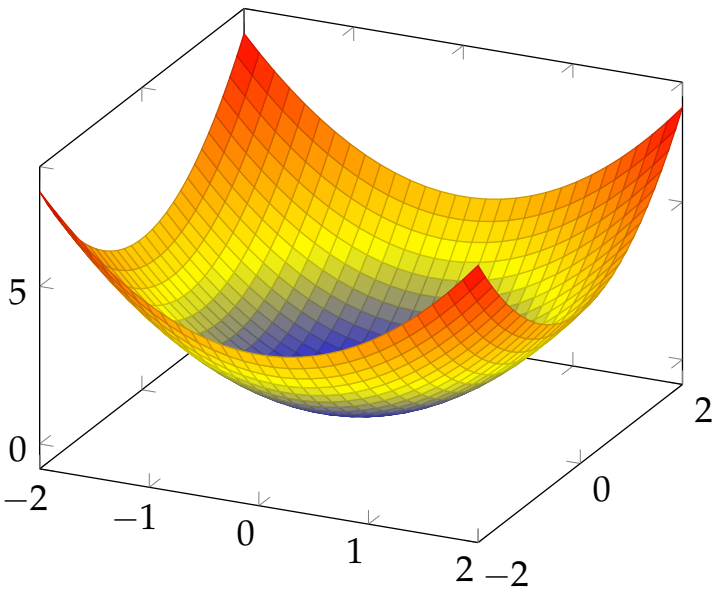
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$$\nabla f = \hat{u}_1 \frac{1}{h_1} \frac{\partial f}{\partial u_1} + \hat{u}_2 \frac{1}{h_2} \frac{\partial f}{\partial u_2} + \hat{u}_3 \frac{1}{h_3} \frac{\partial f}{\partial u_3}$$

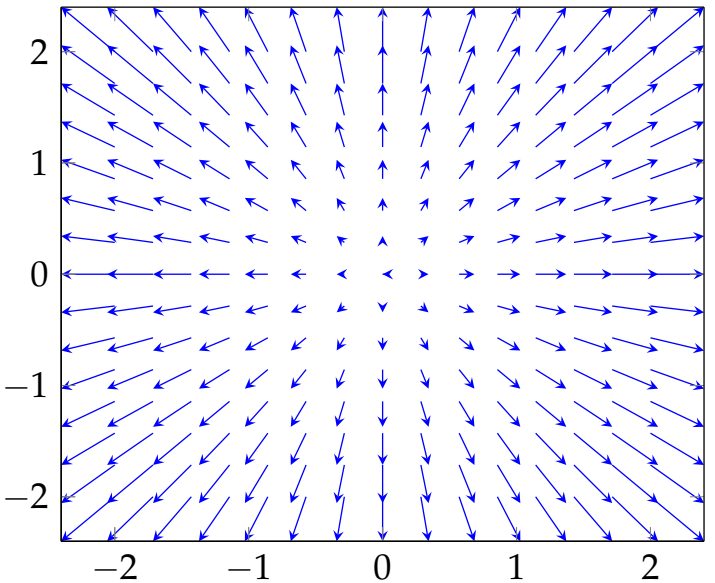
- $\nabla f(a, b)$ points in the direction of greatest increase of f at (a, b)
- $|\nabla f(a, b)|$ is the maximum rate of change of f at (a, b)
- ∇f is perpendicular to the level curves of f (two variable) or level surfaces of f (three variables)



Gradient - Examples



$\nabla f = \hat{x}2x + \hat{y}2y$



Gradient - Examples

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- Find the gradient of $f(x, y, z) = x^4 + y^4$
- Find the gradient of $f(x, y, z) = xy$ at the point (-4,5)

Curl

$\vec{F} = \hat{r}F_r + \hat{\theta}F_\theta + \hat{\phi}F_\phi$

$d\vec{\ell} = \hat{r}\,dr + \hat{\theta}\,r\,d\theta + \hat{\phi}\,r\sin\theta\,d\phi$

$\hat{u}_1 h_1 du_1 + \hat{u}_2 h_2 du_2 + \hat{u}_3 h_3 du_3$

$$\nabla \times F = \begin{vmatrix} \frac{\hat{u}_1}{h_2 h_3} & \frac{\hat{u}_2}{h_1 h_3} & \frac{\hat{u}_3}{h_1 h_2} \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 F_1 & h_2 F_2 & h_3 F_3 \end{vmatrix}$$

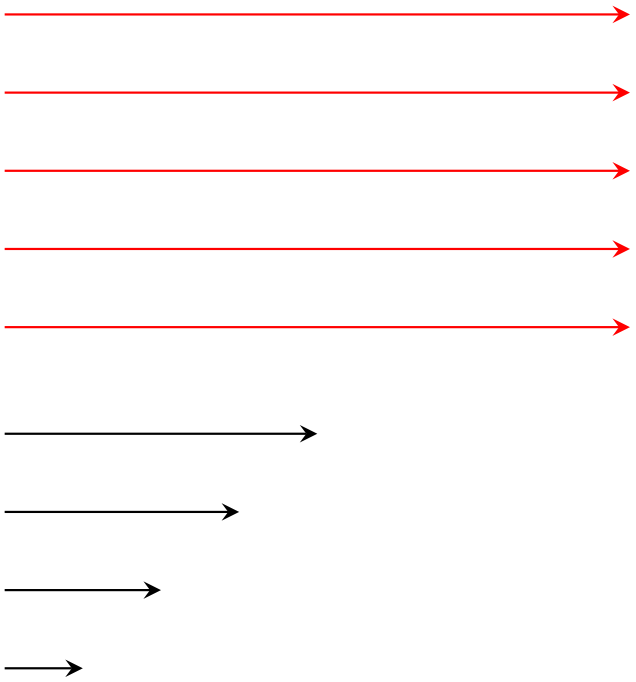
$$\begin{vmatrix} \frac{\hat{r}}{r^2\sin\theta} & \frac{\hat{\theta}}{r\sin\theta} & \frac{\hat{\phi}}{r} \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial\theta} & \frac{\partial}{\partial\phi} \\ F_r & rF_\theta & r\sin\theta F_\phi \end{vmatrix} = \frac{\hat{r}}{r^2\sin\theta}\left[\frac{\partial}{\partial\theta} r\sin\theta F_\phi - \frac{\partial}{\partial\phi} rF_\theta\right]$$

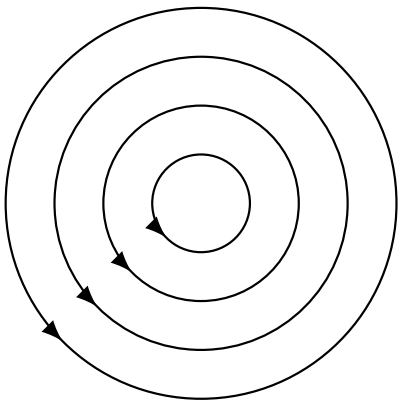
$\sim$

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Curl - Examples

Special case $\nabla \times \vec{H} = 0$ conservative Field





Determine the rotation in the field:

$$\vec{H} = \hat{\mathbf{x}}\left(3y + \frac{z^2}{2}\right) + \hat{\mathbf{y}}\left(2x - z\right) + \hat{\mathbf{z}}3xy$$

Divergence

$$\vec{f} = \hat{u}_1 f_1 + \hat{u}_2 f_2 + \hat{u}_3 f_3$$

$$\nabla \cdot \vec{f} = \frac{1}{h_1 h_2 h_3}\left[\frac{1}{h_1}\frac{\partial h_2 h_3 f_1}{\partial u_1} + \frac{1}{h_2}\frac{\partial h_1 h_3 f_2}{\partial u_2} + \frac{1}{h_3}\frac{\partial h_1 h_2 f_3}{\partial u_3}\right]$$

$h_1 = 1 \quad h_2 = \rho \quad h_3 = 1$

$f_1 = f_\rho \quad f_2 = f_\phi \quad f_3 = z$

$$\nabla \cdot \vec{f} = \frac{1}{\rho}\left[\frac{1}{1}\frac{\partial}{\partial \rho}(\rho(1) f_\rho) + \frac{1}{\rho}\frac{\partial}{\partial \phi}(1)(1) f_\phi + \ldots\right]$$

$f_\rho(\rho, \phi, z)$

Divergence - Example

Determine the electric flux through a point (x, y, z) of the field:

$$\vec{H} = \hat{\mathbf{x}}\left(3y + \frac{z^2}{2}\right) + \hat{\mathbf{y}}\,(2x - z) + \hat{\mathbf{z}}3xy$$

rect: $h_1 = 1$, $h_2 = 1$, $h_3 = 1$

$$\nabla \cdot \vec{H} = \frac{\partial}{\partial x} H_x + \frac{\partial}{\partial y} H_y + \frac{\partial}{\partial z} H_z$$

$$= \frac{\partial}{\partial x}\left(3y + \frac{z^2}{2}\right) + \frac{\partial}{\partial y}(2x - z) + \frac{\partial}{\partial z} 3xy$$

$\frac{\partial}{\partial x}\left(3y + \frac{z^2}{2}\right) \rightarrow 0$, $\frac{\partial}{\partial y}(2x - z) \rightarrow 0$, $\frac{\partial}{\partial z} 3xy \rightarrow 0$

What is Divergence, really?

- Amount of a field "leaving" an enclosed surface:

$$\nabla \cdot \vec{F} = \lim_{\Delta v \to 0} \frac{\oiint_S \vec{F} \cdot d\vec{S}}{\Delta v}$$

- Gauss' Law:

$$\oiint_S \vec{D} \cdot d\vec{S} = \oiint_S \epsilon \vec{E} \cdot d\vec{S} = Q_{\text{enclosed}} \qquad \text{and} \qquad \nabla \cdot \vec{D} = \rho_v$$

Laplacian

$\nabla^2 V = \nabla \cdot \nabla V$

$$\nabla^2 V(x,y,z) = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\nabla^2 V(\rho,\phi,z) = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial V}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

$$\nabla^2 V(r,\theta,\phi) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial V}{\partial r}\right)$$

$$+ \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial V}{\partial \theta}\right) + \frac{1}{r^2 \sin\theta^2}\frac{\partial^2 V}{\partial \phi^2}$$

$\nabla^2 V = \frac{\rho}{\epsilon}$ => Poisson

$\nabla^2 V = 0$ => Laplace's Eq.

Laplacian Examples

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- Find the Laplacian of $f(x,y,z) = e^{xyz}$
- Find the Laplacian of $f(x,y,z) = \dfrac{1}{x^2+y^2+z^2}$

Divergence Theorem & Stoke's Theorem

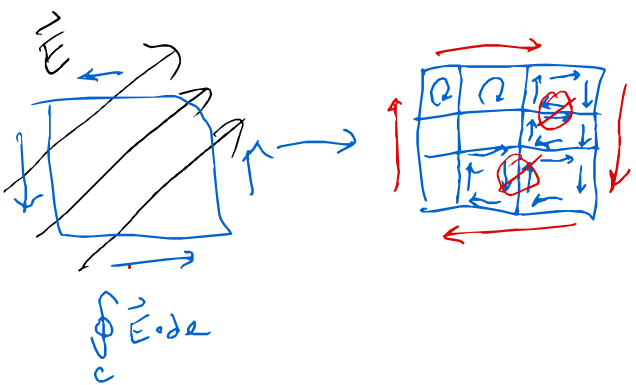
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Divergence Theorem

$$\iiint_V \nabla \cdot \vec{D} dV = \oiint_S \vec{D} \cdot d\vec{S}$$

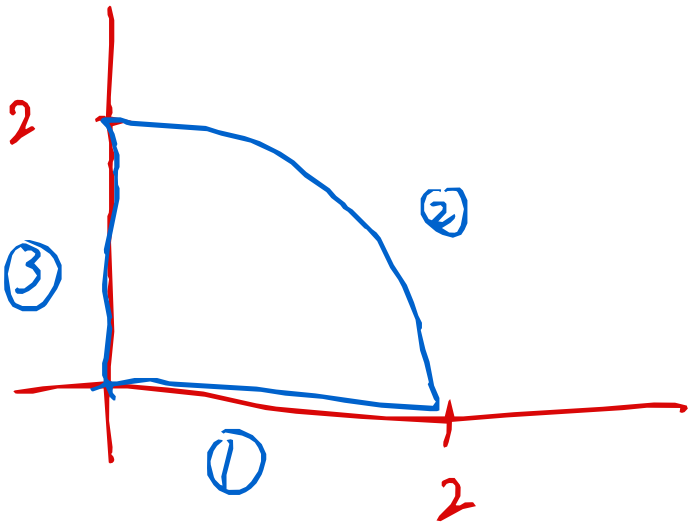
Stoke's Theorem

$$\iint_S (\nabla \times F) \cdot \vec{dS} = \oint_C F \cdot d\ell$$



Stoke's Theorem

Verify the contour integral, using Stoke's theorem:
$\vec{E} = \hat{\rho}\rho\sin\phi + \hat{\phi}\rho^2$ over the contour
$(0,0,0) - (2,0,0) - (2,\frac{\pi}{2},0) - (0,\frac{\pi}{2},0)$



$$\oint_C \vec{E}\cdot d\vec{\ell} = \int_1 + \int_2 + \int_3 \overset{\text{Stoke's}}{=} \iint_S (\nabla\times\vec{E})\cdot\hat{z}\, dS$$